

Characterization of BLUE (Best Linear Unbiased Estimator) under G-M framework

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LSM

$$Y = X\beta + \epsilon$$

$$\epsilon \sim (0, \sigma^2 I)$$

Defⁿ: (Linear Zero function)

A linear zero function (LZF) is a random variable of the form $\underline{c}^T Y$ s.t.

$$\mathbb{E}(\underline{c}^T Y) = 0 \text{ for all } (\beta, \sigma^2) \longrightarrow (*)$$

i.e. $\underline{c}^T Y$ is an unbiased estimator of zero.

Condition (*) means,

$$0 = \mathbb{E}(\underline{c}^T Y) = \underline{c}^T \mathbb{E} Y = \underline{c}^T X \beta \quad \forall \beta$$

$$\Rightarrow \underline{c}^T X = 0 \text{ i.e. } \underline{c} \in \mathcal{C}(X)^\perp$$

$$\exists \underline{c} \neq 0 \text{ if } \text{rank } X < p$$

Theorem: An LUE (Linear Unbiased Estimator) $\underline{l}^T Y$ is a BLUE if and only if $\underline{l}^T Y$ is uncorrelated with each LZF $\underline{c}^T Y$

pf: Suppose the condition holds. Let $\underline{m}^T Y$ be an arbitrary LUE for $\mathbb{E}(\underline{l}^T Y) = \underline{a}^T \beta$ say.

$$\begin{aligned} \text{Then } \text{Var}(\underline{m}^T Y) &= \text{Var}((\underline{m}^T Y - \underline{l}^T Y) + \underline{l}^T Y) \\ &= \text{Var}(\underline{m}^T Y - \underline{l}^T Y) + \text{Var}(\underline{l}^T Y) \\ &\quad + 2\text{Cov}(\underline{m}^T Y - \underline{l}^T Y, \underline{l}^T Y) \end{aligned}$$

$$\text{Since } \mathbb{E}(\underline{m}^T Y - \underline{l}^T Y) = \underline{a}^T \beta - \underline{a}^T \beta = 0$$

i.e. $m^T Y - l^T Y$ is an LFF, by the condition, the covariance term vanishes.

For the converse,
assume that $l^T Y$ is a BLUE.

Let $c^T Y$ be an arbitrary LFF
Consider the family of LUE

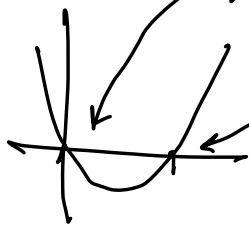
$$l^T Y + \lambda c^T Y \text{ for } \lambda \in \mathbb{R}$$

Since $l^T Y$ is a BLUE

$$\begin{aligned} \text{Var}(l^T Y + \lambda c^T Y) &\geq \text{Var}(l^T Y) \\ \text{or } \text{Var}(l^T Y) + \lambda^2 \text{Var}(c^T Y) + 2\lambda \text{Cov}(l^T Y, c^T Y) &\geq \text{Var}(l^T Y) \end{aligned}$$

$$\text{Solve for } \lambda^2 \text{Var}(c^T Y) + 2\lambda \text{Cov}(l^T Y, c^T Y) = 0$$

$$\lambda = 0, \quad - \frac{2 \text{Cov}(l^T Y, c^T Y)}{\text{Var}(c^T Y)}$$



The only possible way that the inequality is satisfied for all λ , is if

$$\text{Cov}(l^T Y, c^T Y) = 0$$

which shows that $l^T Y$ is uncorrelated with an arbitrary LFF.

Theorem:

Suppose $\underline{a}^T \underline{\beta}$ is estimable. Then there exists a unique BLUE of $\underline{a}^T \underline{\beta}$ and it is of the form $\underline{a}^T \hat{\underline{\beta}}$ where $\hat{\underline{\beta}}$ is any least squares estimator of $\underline{\beta}$.

Pf: First check that $\underline{a}^T \hat{\underline{\beta}}$ is an LUE of $\underline{a}^T \underline{\beta}$

$$\hat{\underline{\beta}} = (\underline{X}^T \underline{X})^{-} \underline{X}^T \underline{Y} \text{ for some } g \text{ inverse } (\underline{X}^T \underline{X})^{-}$$

$$\underline{a}^T \underline{\beta} \text{ estimable} \iff \underline{a} \in \mathcal{R}(\underline{X}) \text{ i.e. } \underline{a} = \underline{X}^T \underline{u} \text{ (say)}$$

$$\underline{a}^T \hat{\underline{\beta}} = \underline{u}^T \underline{X} \hat{\underline{\beta}} = \underline{u}^T \underline{P}_{\underline{X}} \underline{Y} \text{ where } \underline{P}_{\underline{X}} \text{ is the orthogonal projection onto } \mathcal{C}(\underline{X})$$

$$E(\underline{a}^T \hat{\underline{\beta}}) = \underline{u}^T \underline{P}_{\underline{X}} E \underline{Y}$$

$$= \underline{u}^T \underline{P}_{\underline{X}} \underline{X} \underline{\beta} = \underline{u}^T \underline{X} \underline{\beta} = \underline{a}^T \underline{\beta}$$

$$\text{Now let } \underline{c}^T \underline{Y} \text{ be an LZF} \Rightarrow \underline{c}^T \underline{X} = 0$$

$$\text{Cor}(\underline{a}^T \hat{\underline{\beta}}, \underline{c}^T \underline{Y}) = \text{Cor}(\underline{u}^T \underline{P}_{\underline{X}} \underline{Y}, \underline{c}^T \underline{Y})$$

$$= \underline{u}^T \underline{P}_{\underline{X}} \text{Var}(\underline{Y}) \underline{c}$$

$$= \sigma^2 \underline{u}^T \underline{P}_{\underline{X}} \underline{c} = 0$$

$$\underline{P}_{\underline{X}} \underline{c} = \underline{X} (\underline{X}^T \underline{X})^{-} \underline{X}^T \underline{c} = \underline{0} \quad \left. \vphantom{\underline{P}_{\underline{X}} \underline{c}} \right\}$$

So by previous theorem, $\underline{a}^T \hat{\underline{\beta}}$ is a BLUE for $\underline{a}^T \underline{\beta}$

Now suppose $m^T Y$ is any (other) BLUE

then $\text{Var}(m^T Y) = \text{Var}(a^T \hat{\beta})$

Let $\underline{l} = P_X \underline{u}$
so that
 $a^T \hat{\beta} = l^T Y$

$$\Rightarrow \text{Var}(m^T Y - l^T Y) + \text{Var}(l^T Y) + 2 \underset{0}{\text{Cov}}(m^T Y - l^T Y, l^T Y) = \text{Var}(l^T Y)$$

$$\Rightarrow m^T Y = l^T Y \text{ a.s.}$$

What if we are estimating a vector-valued parameter

$$\theta = A\beta$$

$m \times p \quad p \times 1$

$$= \begin{pmatrix} a_1^T \beta \\ \vdots \\ a_m^T \beta \end{pmatrix}$$

Note $A\beta$ is estimable, if there exists L s.t.

$$A\beta = E(L^T Y) \text{ for all } \beta$$

i.e. $A = L^T X$

Then $A\beta$ is estimable, if and only if

$a_j^T \beta$ is estimable for each $j = 1, \dots, m$

Suppose we choose the L^2 -loss defined as

$$L(\hat{\theta}, \theta) = \|\hat{\theta} - \theta\|_2^2$$

$$\begin{aligned} E(\|\hat{\theta} - \theta\|_2^2) &= E\|\hat{\theta} - E\hat{\theta} + (E\hat{\theta} - \theta)\|_2^2 \\ &= E\|\hat{\theta} - E\hat{\theta}\|_2^2 + \underbrace{E\|E\hat{\theta} - \theta\|_2^2}_{\text{Bias}_0(\hat{\theta})} \end{aligned}$$

$$= \mathbb{E}\{\text{Trace}[(\hat{\theta} - \mathbb{E}\hat{\theta})(\hat{\theta} - \mathbb{E}\hat{\theta})^T]\} + \mathbb{E}\|\mathbb{E}\hat{\theta} - \theta\|_2^2$$

$$= \text{Trace}(\text{Var } \hat{\theta}) + \|\text{Bias } \hat{\theta}\|_2^2$$

So the BLUE of $A\beta$ (in the sense of minimizing the l^2 risk among all LUE) is $A\hat{\beta}$, where $\hat{\beta}$ is any LSE

$\text{Trace}(\text{Var } \hat{\theta}) = \sum_{j=1}^m \text{Var } \hat{\theta}_j \leftarrow$ minimize each coordinate within the class of linear unbiased estimators for the corresponding parameter $\theta_j = a_j^T \beta$

The corr. BLUE is $a_j^T \hat{\beta}$ as a consequence

$\text{Trace}(\text{Var } \hat{\theta})$ is minimized within the class of LUE for $A\beta$ by $\hat{\theta} = A\hat{\beta}$

Ex 1. $Y = \beta_1 X_{1i} + \dots + \beta_p X_{pi} + \epsilon_i$

Estimation: $A\beta \leftarrow$ for some matrix A

If $X^T X$ is nonsingular, i.e. X has full column rank, then $A\beta$ is always estimable and then the BLUE is $A(X^T X)^{-1} X^T Y$.

Ex 2. $Y_{ij} = \alpha_i + \gamma_j + \epsilon_{ij}$

$$\begin{pmatrix} Y_{11} \\ Y_{12} \\ Y_{21} \\ Y_{22} \end{pmatrix} = \underbrace{\begin{pmatrix} 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{pmatrix}}_X \begin{pmatrix} \alpha_1 \\ \alpha_2 \\ \gamma_1 \\ \gamma_2 \end{pmatrix} + \begin{pmatrix} \epsilon_{11} \\ \epsilon_{12} \\ \epsilon_{21} \\ \epsilon_{22} \end{pmatrix}$$

$$\text{rank}(X) = 3, \quad R(X) = \{(\gamma_1, \gamma_2, \gamma_3, \gamma_4) : \gamma_1 + \gamma_2 = \gamma_3 + \gamma_4\}$$

Ex. Which of the following parameters are estimable

a) α_1

$$a^T = (1 \ 0 \ 0 \ 0) \quad X$$

b) $\gamma_1 + \gamma_2$

$$a^T = (0 \ 0 \ 1 \ 1) \quad X$$

c) $\alpha_1 + \gamma_1$

$$a^T = (1 \ 0 \ 1 \ 0) \quad \checkmark$$

d) $\alpha_1 + \alpha_2 - \gamma_1 - \gamma_2$ $a^T = (1 \ 1 \ -1 \ -1) \quad X$

e) $(\alpha_1 - \alpha_2) + (\gamma_1 - \gamma_2)$ $a^T = (1 \ -1 \ 1 \ -1) \quad \checkmark$

$$\beta = \begin{pmatrix} \alpha_1 \\ \alpha_2 \\ \gamma_1 \\ \gamma_2 \end{pmatrix}$$

Define $\mu_{ij} = \alpha_i + \gamma_j \quad \begin{matrix} i=1,2 \\ j=1,2 \end{matrix}$

$$Y_{ij} = \mu_{ij} + \epsilon_{ij} \quad \text{where } \mu_{ij} = \alpha_i + \gamma_j$$

$$\underline{Y} = \underline{\mu} + \underline{\epsilon} \quad \leftarrow \text{LS est. of } \underline{\mu} \text{ is } \underline{Y}$$

$$\alpha_1 + \gamma_1 = \mu_{11} = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}^T \underline{\mu}$$

$$(\alpha_1 - \alpha_2) + (\gamma_1 - \gamma_2) = (\alpha_1 + \gamma_1) - (\alpha_2 + \gamma_2)$$

$$= \mu_{11} - \mu_{22}$$

$$= \begin{pmatrix} 1 \\ 0 \\ 0 \\ -1 \end{pmatrix}^T \underline{\mu}$$